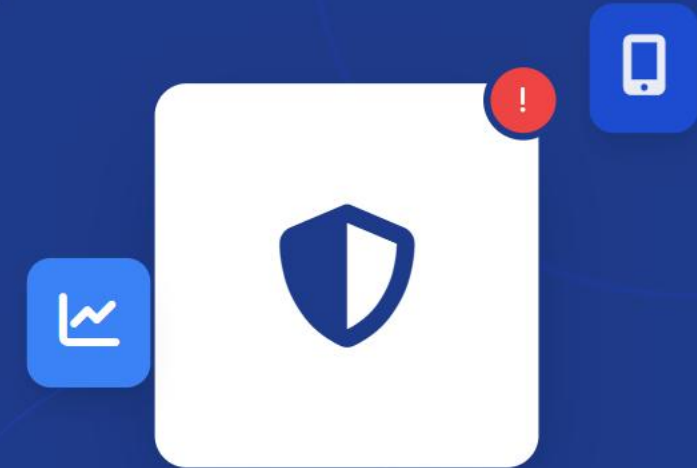


Digital Incident & Near-Miss Reporting System

| Project Overview Presentation

PROJECT TEAM

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Safety • Compliance • Analytics

Problem Overview

The current manual safety management landscape faces critical inefficiencies affecting risk mitigation and compliance.



Safety Risks

Inadequate reporting mechanisms cause underreported incidents and near-misses. This lack of visibility directly increases workplace accidents and compromises overall personnel safety.



Manual Limitations

Reliance on paper-based forms creates time-consuming and error-prone processes. Physical paperwork delays information flow, slowing down response times and corrective actions.



Compliance Issues

Inconsistent documentation and fragmented records lead to serious audit failures. The lack of a centralized digital trail creates regulatory risks and potential legal liabilities.

Our Solution: Digital Safety Platform

A comprehensive digital approach designed to streamline reporting, accelerate response times, and ensure regulatory compliance through three core pillars.



Smart Reporting

An intuitive mobile-based interface allows employees to report incidents instantly. Features include photo capture, location tagging, and detailed voice or text descriptions for ease of use.



Proactive Alerts

Automated workflows trigger instant notifications to supervisors immediately after a report is filed. This ensures rapid assessment, severity classification, and quick corrective action.

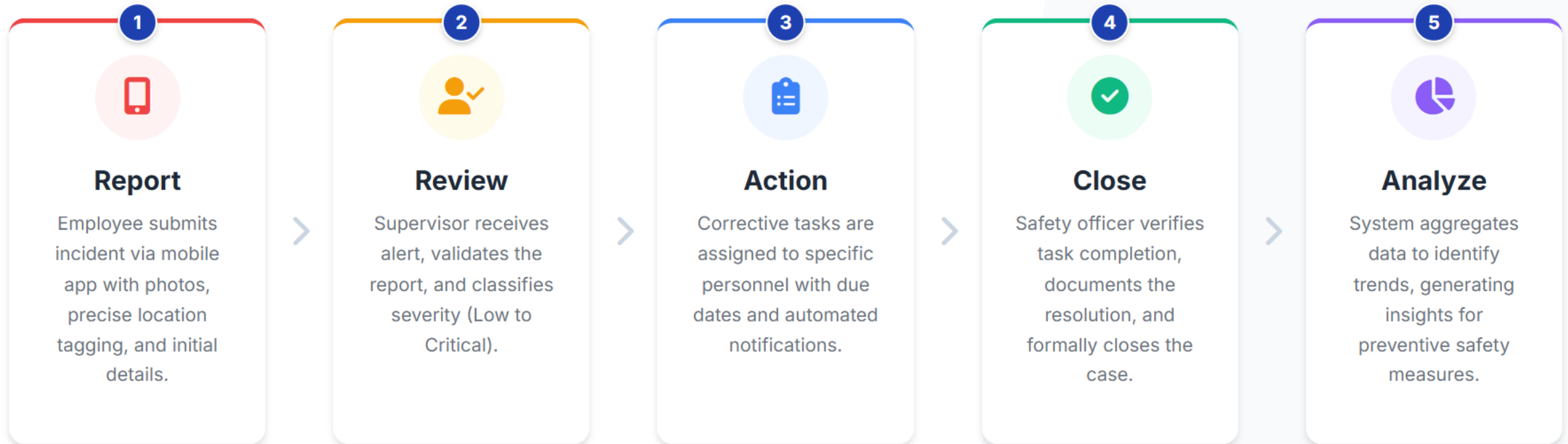


Centralized Records

A secure, cloud-based digital repository consolidates all safety data. This enables powerful trend analysis, simplifies audit readiness, and creates a permanent, searchable safety trail.

Process Flow: From Incident to Resolution

A streamlined end-to-end digital process ensuring every reported issue is tracked, addressed, and analyzed for future prevention.



User-Friendly Incident Reporting

Empowering every employee to contribute to safety with a simple, intuitive mobile interface designed for rapid field use.



Quick Mobile Submission

Streamlined form designed for completion in under 60 seconds.



Photo Capture & Upload

Direct integration with device camera for evidence collection.



Automatic Location Tagging

GPS coordinates automatically attached to every report.



Detailed Incident Description

Voice-to-text and structured fields for accurate details.



Real-Time Submission Status

Instant confirmation and tracking number generation.

The smartphone screen displays the 'New Report' form. At the top, there is a blue header with a hamburger menu icon on the left and a bell icon on the right. Below the header, the title 'New Report' is prominently displayed in white, with 'Incident Reporting System' in a smaller font underneath. The form is divided into several sections: 1. 'LOCATION' section with a red location pin icon, the text 'Warehouse Zone B, Aisle 4', and a small edit icon. 2. 'INCIDENT TYPE' section with a warning triangle icon, the text 'Near Miss - Forklift', and a dropdown arrow. 3. 'EVIDENCE' section featuring a dashed box containing a camera icon and the text 'Tap to take photo'. At the bottom of the screen, there are two grey rectangular buttons.

Supervisor Dashboard & Actions

A centralized command center providing real-time oversight and tools for rapid incident resolution.



Real-time Alerts

Instant push notifications for new incidents.

New Incident #402

2m ago

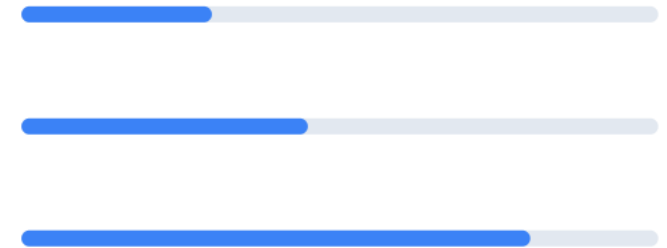
Chemical spill reported in Lab 3.



Report Validation

Review evidence and verify incident details before processing.

- ✓ Evidence Verified
- ✓ Location Confirmed



Severity Classification

One-click categorization system.

CRITICAL

HIGH

MEDIUM

LOW

DIGITAL INCIDENT REPORTING SYSTEM



Assign Actions

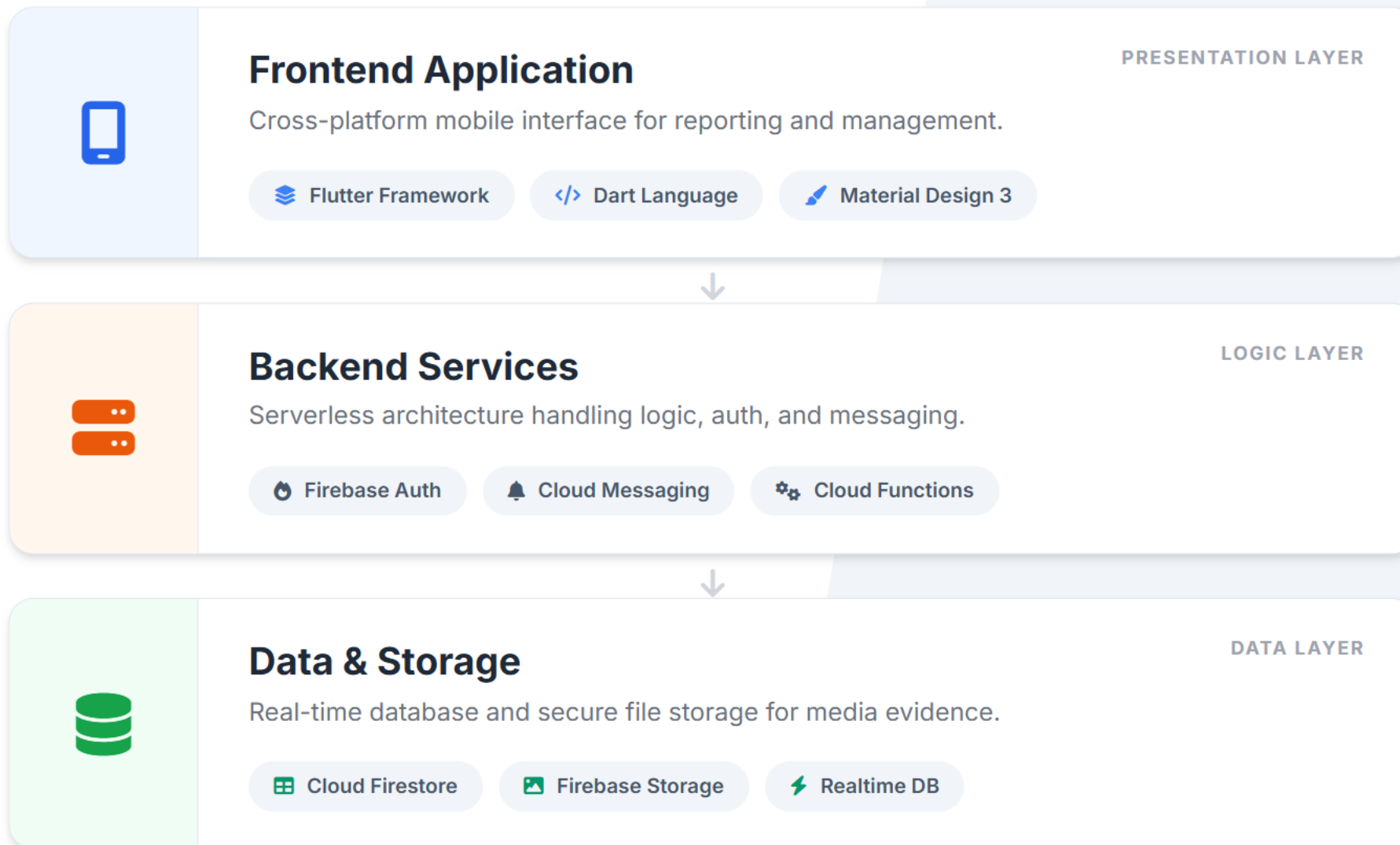
Delegate corrective tasks to maintenance or safety teams.



[Go to Analytics](#)

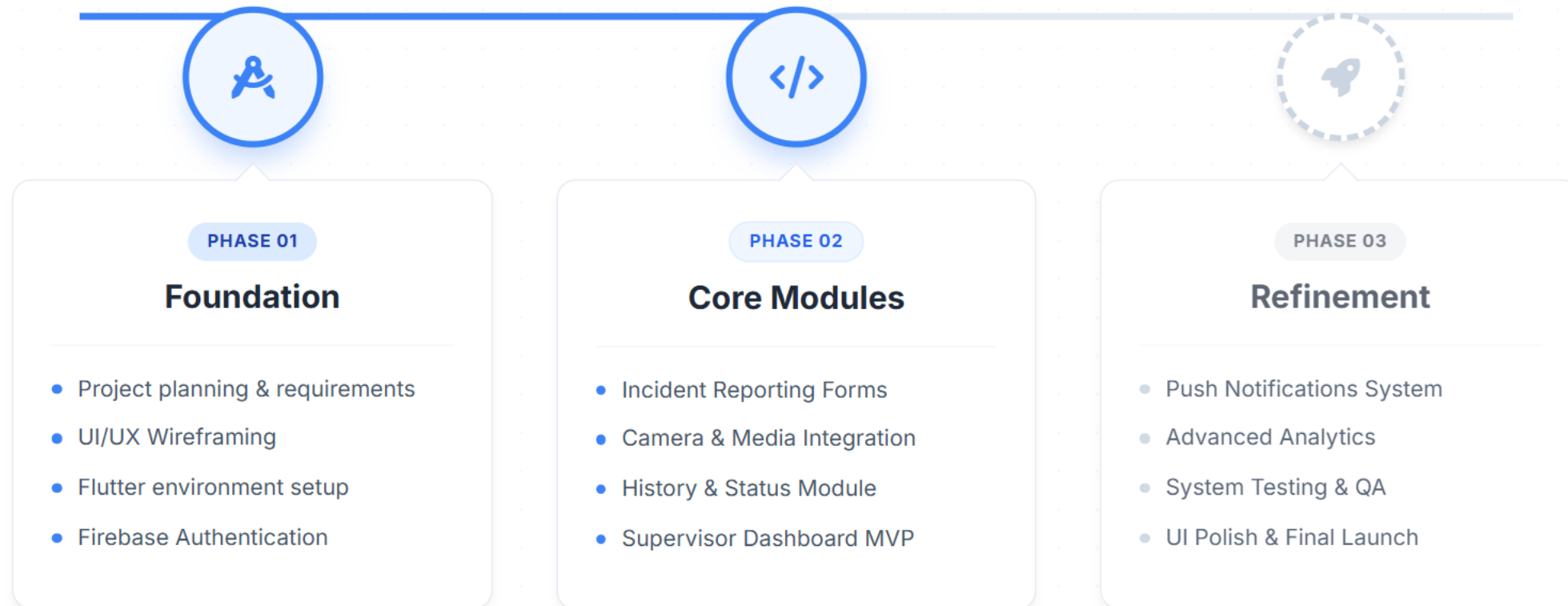
Technology Stack

A robust, scalable architecture leveraging modern cross-platform frameworks and cloud-native backend services.



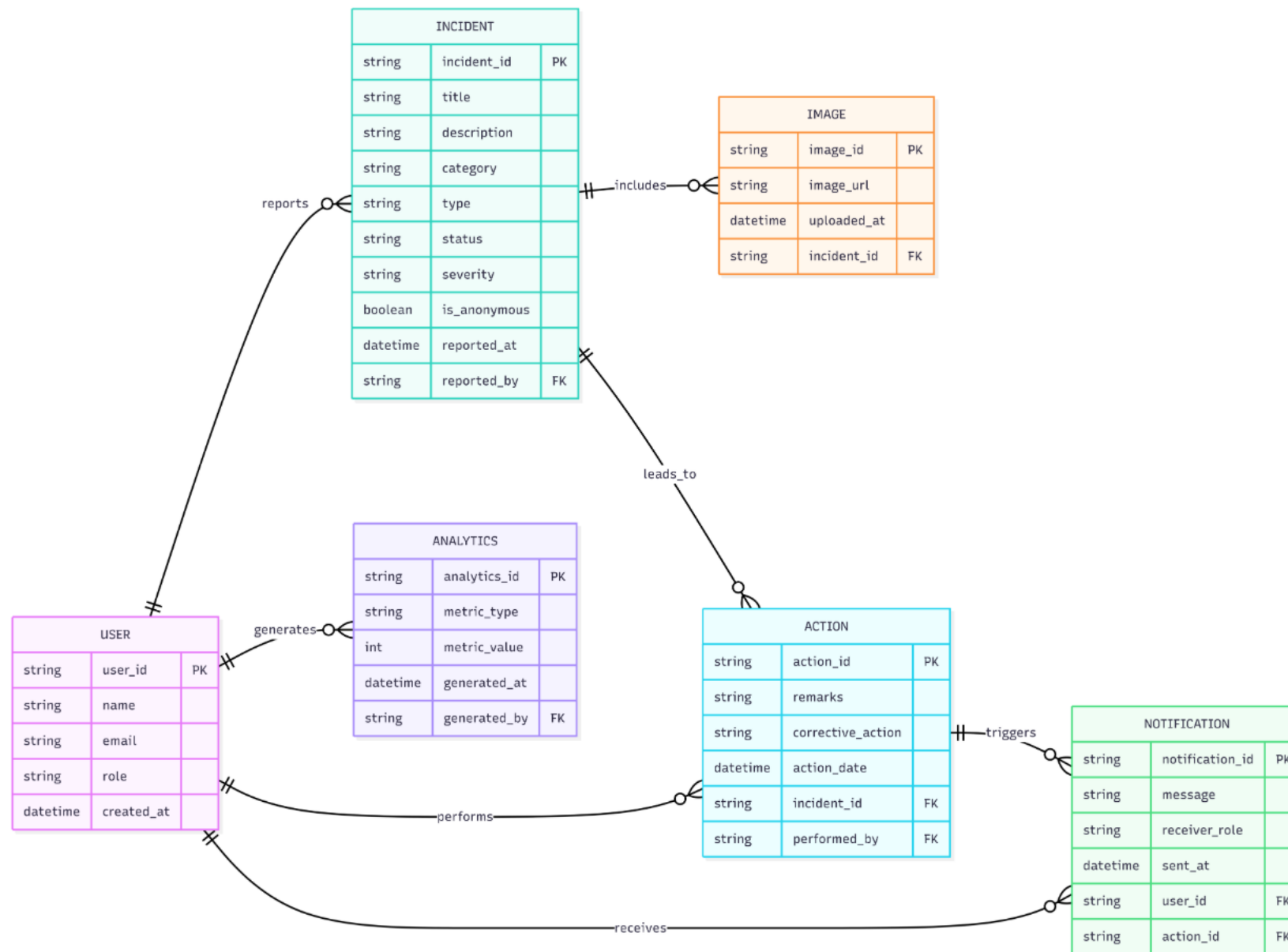
Planned Development Timeline

A structured three-phase approach to delivering a comprehensive and compliant digital safety platform.



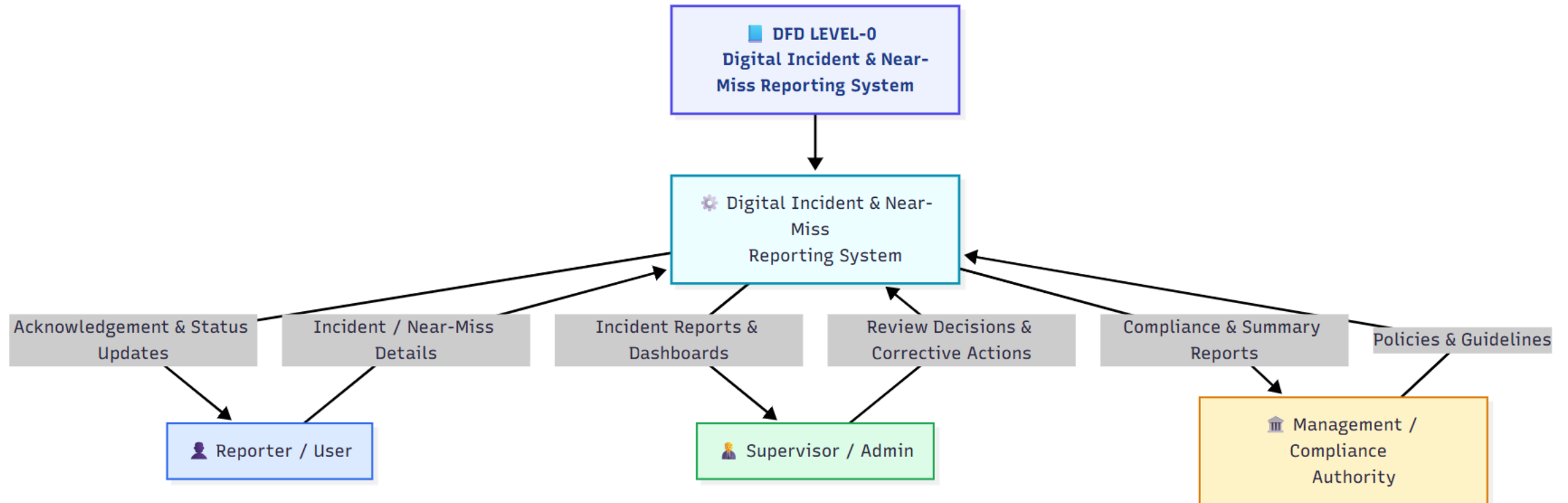
Database Design - ER Diagram

Visual representation of the database schema, entity relationships, and data structure constraints.



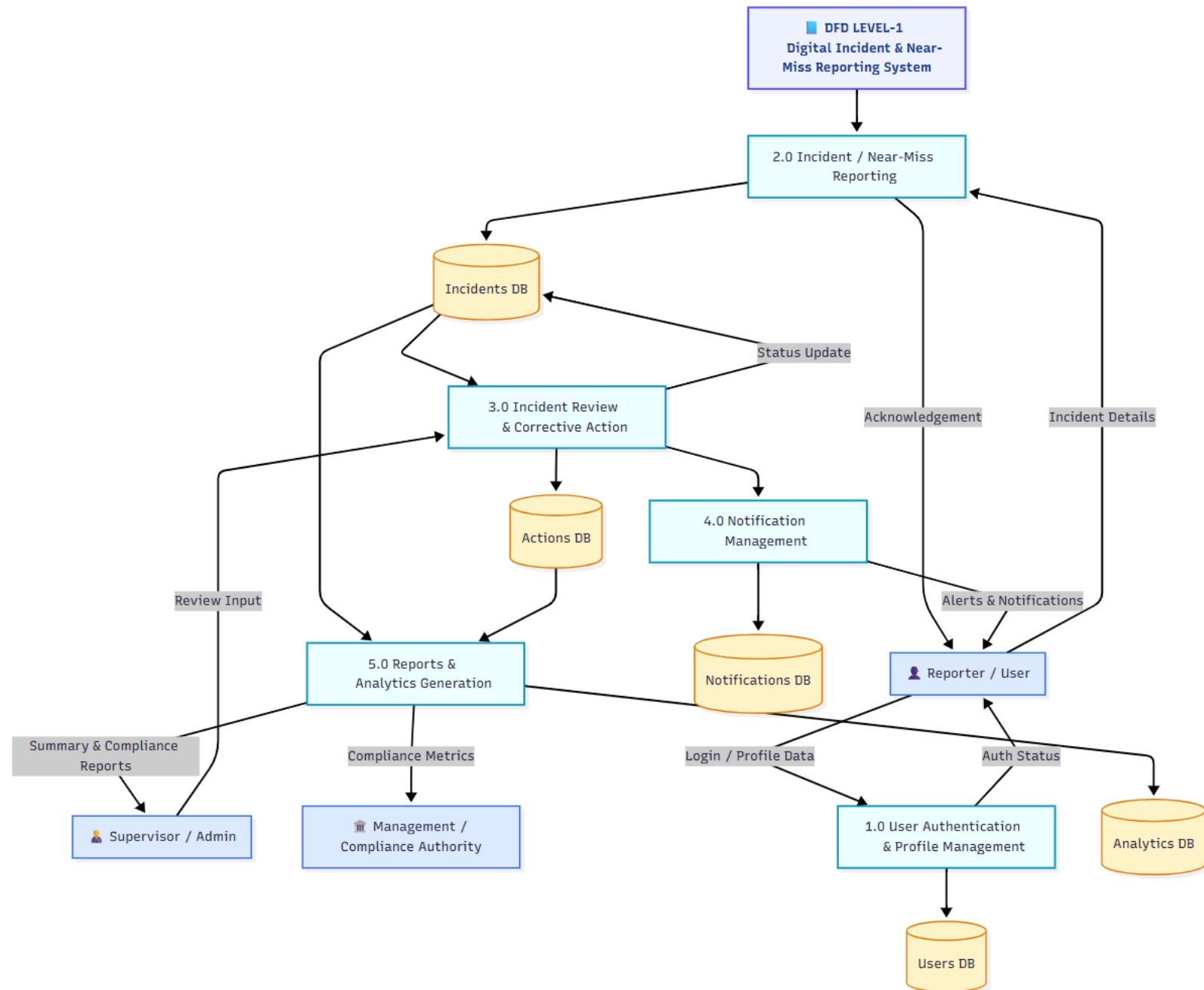
System Design - Data Flow Diagrams

Visual representation of information flow, process transformations, and data storage within the reporting system.



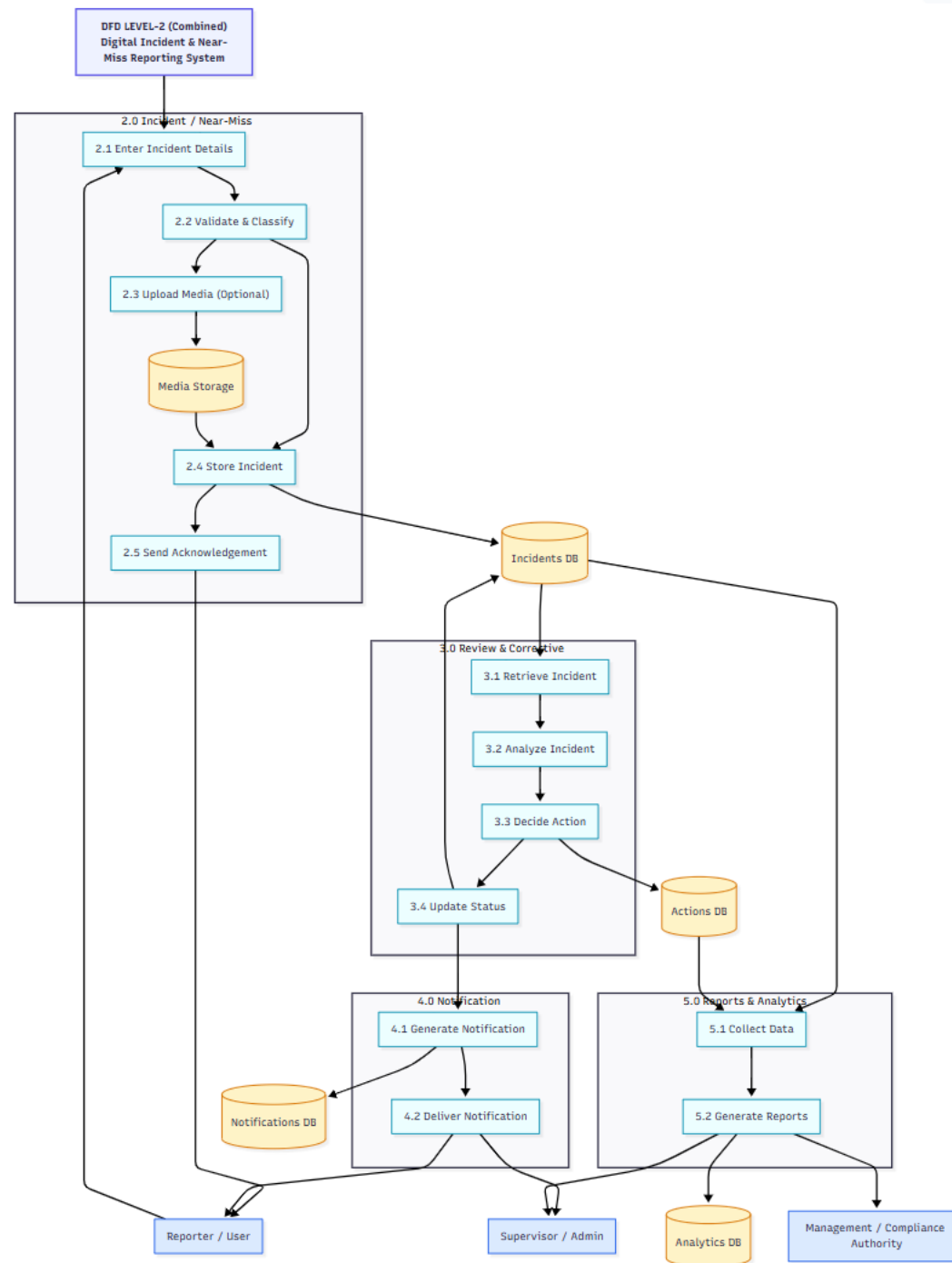
System Design - Data Flow Diagrams

The system captures incident data from users, processes it through review and action workflows, notifies relevant stakeholders, and generates analytical reports for compliance and decision-making.



System Design - Data Flow Diagrams

This diagram presents a consolidated view of all Level-2 processes, showing how incident reporting, review, notification, and analytics workflows interact within the system. It provides an integrated understanding of data flow across the entire lifecycle of an incident.



Testing & Security Strategy

Ensuring system reliability and data integrity through comprehensive validation protocols and robust security measures.



Testing Plans

Validation & Performance



Lifecycle & Multi-device Testing

Verifying app stability across different Android versions and device screen sizes to ensure consistent performance.



Edge Case Handling

Stress-testing network interruptions, offline data submission, and handling invalid inputs gracefully.



UI & Usability Testing

Conducting user flow validation to minimize friction in incident reporting and dashboard navigation.



Security Plans

Data Protection & Access Control



Role-Based Authentication

Implementing Firebase Auth to strictly separate employee (reporter) and supervisor (admin) privileges.



Firebase Security Rules

Server-side logic to validate read/write requests, ensuring users can only modify their own reports unless authorized.



Secure Data & Image Handling

Encrypted storage for sensitive incident photos and personal user data within Firebase Storage.

Future Scope & Enhancements

Strategic scalability features planned for post-MVP implementation to enhance data intelligence and organizational reach.



Advanced Analytics & Risk Prediction HIGH PRIORITY

Implementation of AI-driven models to predict potential incident hotspots based on historical data patterns and seasonality.



Anonymous & Voice-Based Reporting HIGH PRIORITY

Privacy-focused submission mode for whistleblowers and voice-to-text integration for hands-free incident logging.



Automated Risk Scoring MEDIUM PRIORITY

Dynamic algorithm to auto-calculate incident severity scores (1-10) based on category, location, and potential impact.



Multi-Organization & Role Expansion MEDIUM PRIORITY

SaaS architecture support for managing multiple distinct client organizations and custom role-based permission sets.



Email & SMS Alert Integration LOW PRIORITY

External notification gateways (Twilio/SendGrid) to push critical alerts to stakeholders outside the mobile app ecosystem.

Expected Benefits & Impact

Measurable improvements in safety culture, operational efficiency, and organizational compliance.



Faster Incident Reporting

Eliminates paper trails, allowing employees to report hazards in seconds via mobile, reducing lag time between occurrence and notification by up to 80%.



Real-Time Alerts

Instant push notifications ensure supervisors and safety officers are immediately aware of critical incidents, enabling rapid response and containment.



Data-Driven Decisions

Transforms raw reports into actionable insights. Trend analysis helps identify recurring root causes and allocate safety resources more effectively.



Improved Compliance Readiness

maintain a digital, audit-ready trail of all incidents and corrective actions. Ensures regulatory adherence to safety standards (OSHA/ISO) and simplifies legal documentation.



Reduced Workplace Risks

Proactive reporting of near-misses allows organizations to fix hazards before they cause harm, leading to a significant long-term reduction in workplace accidents and injuries.

Conclusion

Recapping the core achievements and the strategic value of the Digital Incident Reporting System.

01



Digital Transformation of Safety

Successfully introduced a modern, digital approach to safety reporting, replacing inefficient manual paper trails with instant mobile data capture.

02



Robust System Design

Established a clear, logical system architecture and workflow that ensures data integrity from incident capture through to resolution and analysis.

03



Foundation for Growth

Built a strong technical foundation using scalable technologies (Flutter, Firebase) ready for future enhancements like AI prediction and advanced analytics.

04



Scalable & Practical Solution

Delivered a practical, user-centric solution that addresses immediate compliance needs while remaining scalable for larger organizational deployments.



Thank You

Questions & Discussion



Abhishek Kumar

TEAM MEMBER



Ajayram Meena

TEAM LEAD

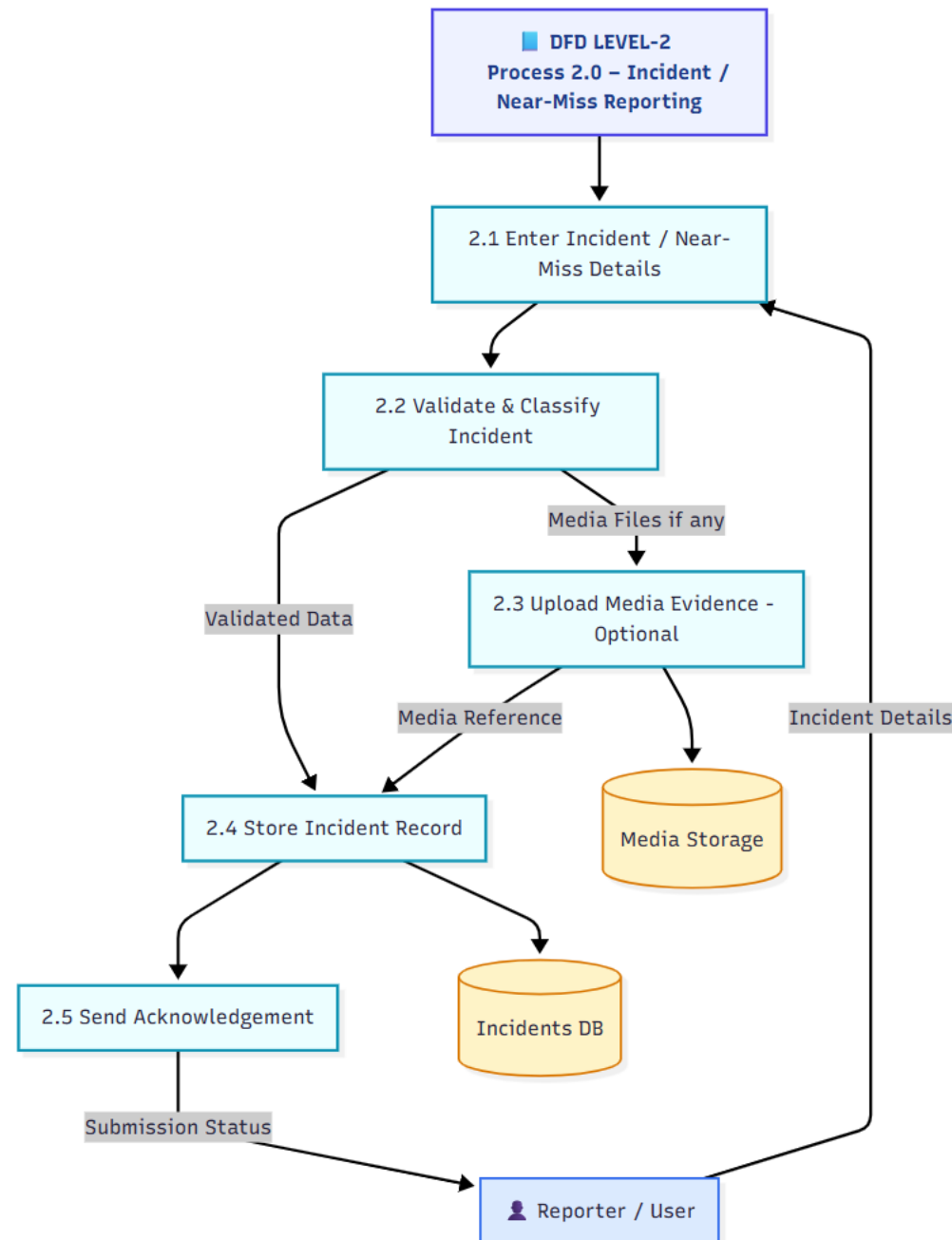


Aniket Kumar Singh

TEAM MEMBER

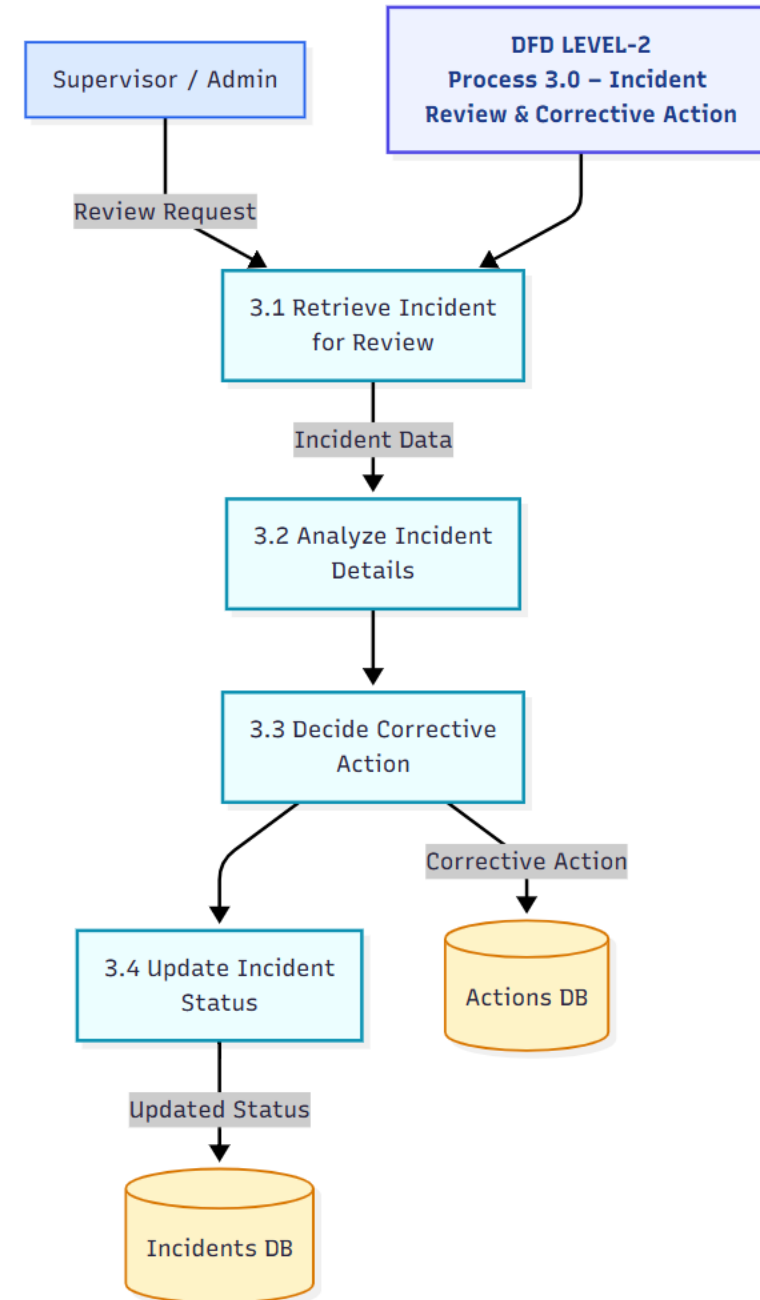
System Design - Data Flow Diagrams

This diagram details how incident data is entered by users, validated, optionally supported with media evidence, and securely stored in the system. An acknowledgement is sent to the user after successful submission.



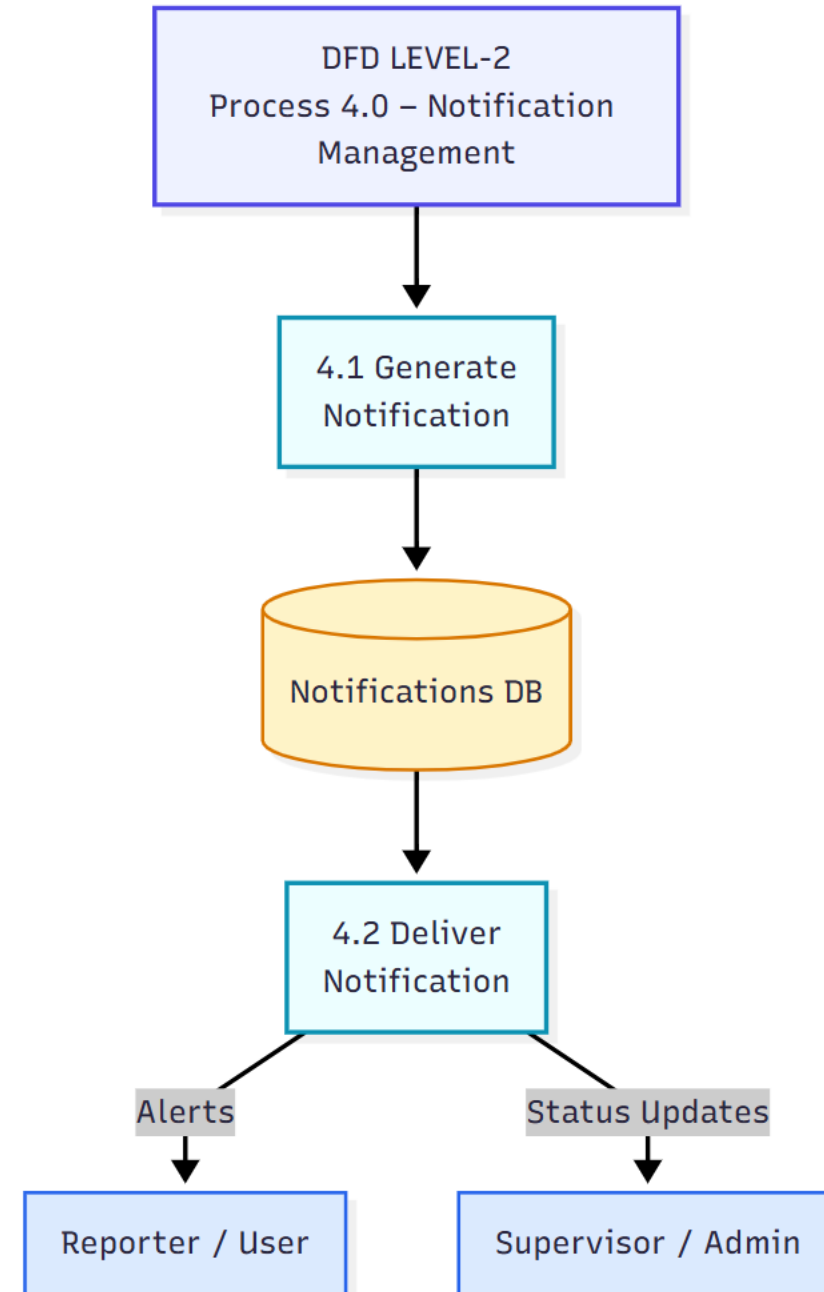
System Design - Data Flow Diagrams

This Level-2 DFD shows how reported incidents are reviewed by supervisors, corrective actions are decided, and the incident status is updated in the system.



System Design - Data Flow Diagrams

This Level-2 DFD describes how system notifications are generated, stored, and delivered to relevant users and administrators to ensure timely alerts and status updates.



System Design - Data Flow Diagrams

This Level-2 DFD explains how incident and corrective action data is analyzed to generate reports and compliance metrics for supervisors and management.

